Fazoviy jismlarning yoyilmasi

Karton qutini qirralari bo'ylab qirqib, stol ustiga tekis yoysak, bir nechta ko'pburchakdan tuzilgan shakl chiqadi. Shu shaklni qayta buklasak, yana o'sha quti hosil bo'ladi.

1-QISM

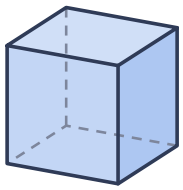
Mavzu

ESLAB OLAYLIK

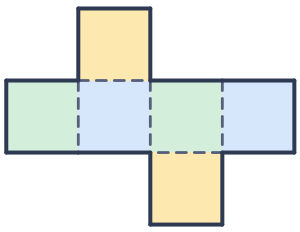
- Kubning 6 ta yog'i bor — hammasi teng kvadratlar. Kuboidning 6 ta yog'i — to'g'ri to'rtburchaklar, qarama-qarshi yoqlari teng. Ikki yoq qirra bo'ylab tutashadi.
- Prizmaning ikki asosi — teng ko'pburchaklar, yon yoqlari — to'g'ri to'rtburchaklar. Piramidaning bitta asosi bor, yon yoqlari — uchi umumiy bo'lgan uchburchaklar.

Kub va kuboidning yoyilmasi

Kub shaklidagi sovg'a qutisini olamiz. Uni ba'zi qirralari bo'ylab qaychi bilan qirqamiz va yoqlarini ochib, stolga tekis yoyamiz.



Kub



Kubning yoyilmasi

USUL · YOYILMA HOSIL QILISH

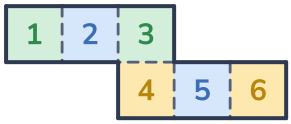
1) Jismni ba'zi qirralari bo'ylab kesamiz. 2) Yoqlarini ochib, tekislikka yoyamiz.

Hosil bo'lgan tekis shakl jismning **yoyilmasi** deyiladi. Yoyilmadagi punktir chiziqlar — buklanadigan joylar.

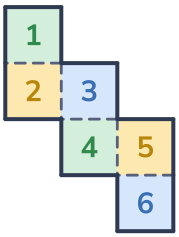
QOIDA

Fazoviy jism yoyilmasini buklab, fazoviy jism hosil qilish mumkin.

1-MISOL · KUBNING YANA IKKI YOYILMASI



Qarama-qarshi yoqlar: 1 va 3, 2 va 5, 4 va 6



Qarama-qarshi yoqlar: 1 va 4, 2 va 5, 3 va 6

Ikkalasi ham 6 ta teng kvadratdan iborat va buklansa, kub hosil bo'ladi. Bir xil rangli kvadratlar kubning qarama-qarshi yoqlari bo'ladi.

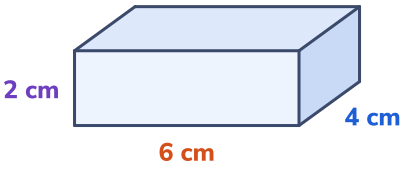
Demak, bitta kubning bir nechta har xil yoyilmasi bo'ladi. Lekin 6 ta kvadratdan tuzilgan har qanday shakl ham kub yoyilmasi bo'lavermaydi.

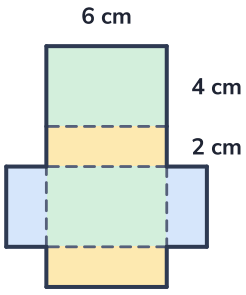
USUL · KUB YOYILMASINI TEKSHIRISH

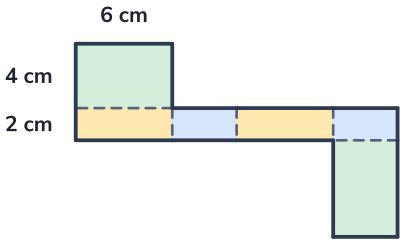
1) Shaklni katakli qog'ozga chizib, qirqib olamiz. 2) Punktir chiziqlar bo'ylab buklaymiz. 3) Kub hosil bo'ldimi?

2-MISOL · KUBOIDNING IKKI XIL YOYILMASI

Bo'yi 6 cm, eni 4 cm, balandligi 2 cm bo'lgan qutini qirqib yoyamiz.







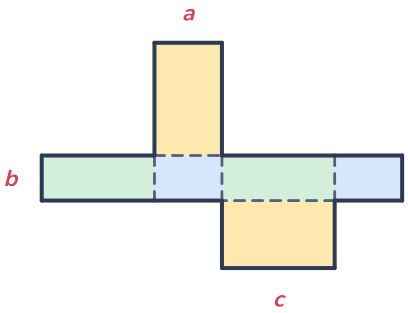
Ikkala yoyilmada ham o'sha 3 juft teng yoq bor: ostki va ustki — 6×4 (yashil), old va orqa — 6×2 (sariq), ikki yon yoq — 4×2 (ko'k). Faqat ular boshqacha joylashgan.

USUL · YOYILMADAGI KESMALAR UZUNLIGI

Yoyilmadagi har bir to'g'ri to'rtburchakning tomonlari — kuboidning ikkita o'lchami. Buklanganda bir-biriga tutashadigan ikki tomon teng bo'ladi.

3-MISOL · A, B, C NI TOPING

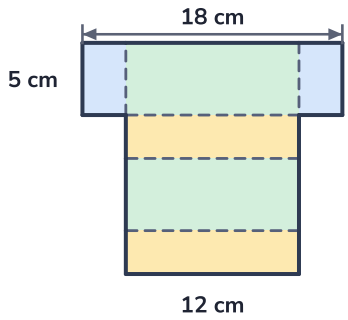
Kuboidning bo'yi 10 cm, eni 4 cm, balandligi 6 cm. Yoyilmadagi a, b, c kesmalar necha cm?



1. Yashil yoqlar — 10×4, sariq — 10×6, ko'k — 4×6.
2. a — sariq yoqning qisqa tomoni: 6 cm; b — yashil yoqning qisqa tomoni: 4 cm; c — sariq yoqning uzun tomoni: 10 cm.

4-MISOL · YOYILMADAN O'LCHAMLARNI TOPISH

Yoyilmaga qarab kuboidning uchta o'lchamini toping.

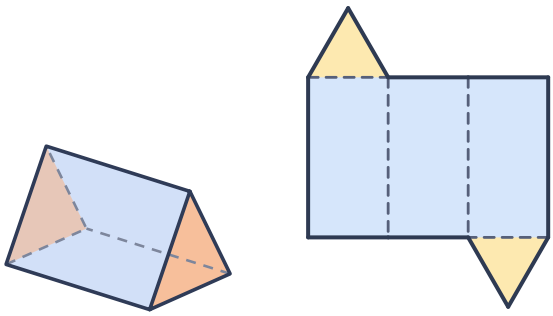


1. Yuqori qator: yon yoq + katta yoq (12 cm) + yon yoq $= 18$ cm. Ikki yon yoq: $18 - 12 = 6$, bittasi $6 : 2 = 3$ cm.
2. Yon yoqning bo'yi 5 cm — katta yoqning ikkinchi tomoni. O'lchamlar: 12, 5, 3 cm.

Prizma va piramidaning yoyilmasi

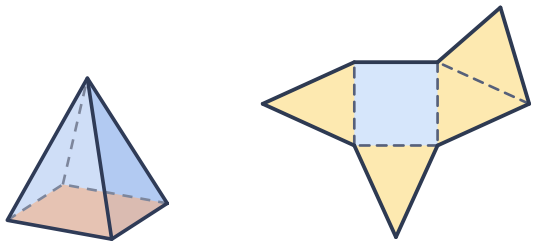
Uchburchakli prizma yoyilmasi 2 ta teng uchburchak (asoslar) va 3 ta to'g'ri to'rtburchakdan (yon yoqlar) iborat. Piramida yoyilmasi asos va uchburchak shaklidagi yon yoqlardan iborat.

5-MISOL · PRIZMANING YOYILMASI



1. Uchta to'g'ri to'rtburchak buklanib, «quvur» bo'ladi. Uchburchaklar uning ikki uchini yopadi: biri tepada, biri pastda.

6-MISOL · ASOSI KVADRAT BO'LGAN PIRAMIDA



1. 1 ta kvadrat (asos) va 4 ta teng uchburchak (yon yoqlar). Bitta uchburchak asosga emas, qo'shni uchburchakka yopishgan — bu ham mumkin.

7-MISOL · ASOSI UCHBURCHAK BO'LGAN PIRAMIDA



4 ta uchburchak: o'rtadagisi — asos, uch tomonidagilari — yon yoqlar. Ular buklanib, uchlari bitta nuqtada uchrashadi.

2-QISM

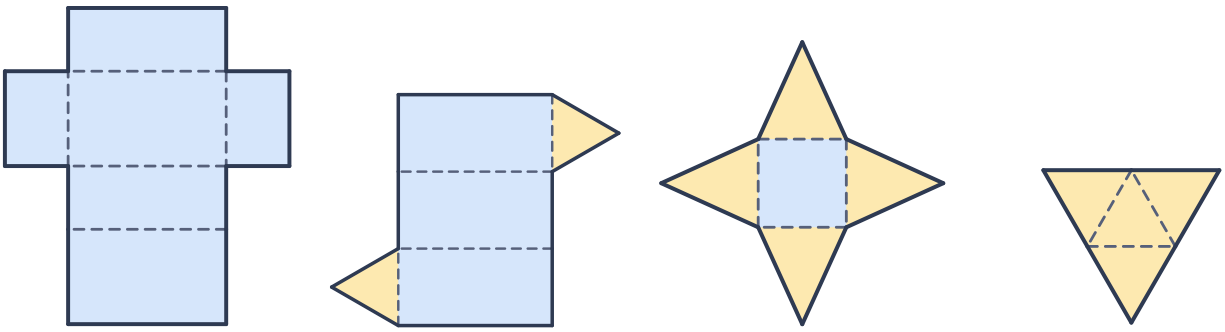
Misollar ishlash



Yoyilmaga qaraganda avval shakllarni sanang: nechta shakl bo'lsa, jismning shuncha yog'i bor.

1-MISOL · YOYILMADAN JISMNI ANIQLASH

Har bir yoyilmadan qanday jism hosil bo'ladi?



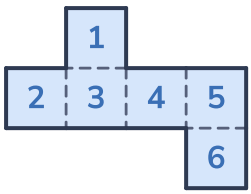
A B C D

1 Yoyilmadagi shakllarni sanaymiz va turini aniqlaymiz: shakllar soni — jismning yoqlari soni.

Yoyilma	Shakllar	Jism
A	6 ta to'g'ri to'rtburchak, 3 juft teng	kuboid
B	2 ta teng uchburchak, 3 ta to'g'ri to'rtburchak	prizma
C	1 ta kvadrat, 4 ta uchburchak	piramida (asosi kvadrat)
D	4 ta uchburchak	piramida (asosi uchburchak)

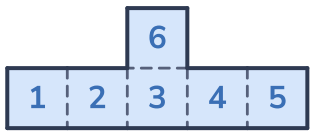
Qo'limizda qog'oz bo'lmasa, shaklni xayolan buklaymiz. Bitta kvadratni kubning **tubi** deb olamiz. Unga yopishgan kvadratlar tepaga buklanib, **devor** bo'ladi. Tubning ro'parasidagi yoq — **qopqoq**. Har kvadrat boshqa-boshqa yoqqa tushsa — kub yoyilmasi; ikki kvadrat ustma-ust tushsa — emas.

2-MISOL · BU KUB YOYILMASIMI?



1. 3 — tub; 2 va 4 — chap va o'ng devor; 1 — orqa devor.
2. 5 o'ng devor ustiga buklanib, qopqoq bo'ladi; 6 qopqoqdan osilib, old devor bo'ladi. Javob: ha.
3. Qarama-qarshi yoqlar: 3 va 5 (tub va qopqoq), 2 va 4, 1 va 6. Qatorda ketma-ket turgan 2, 3, 4 ning chetdagilari — 2 va 4 — qarama-qarshi chiqdi.

3-MISOL · BU KUB YOYILMASIMI?



1. 2 — tub; 1 va 3 — chap va o'ng devor; 6 — orqa devor; 4 — qopqoq.
2. 5 qopqoqdan osilib, chap devorga — 1 turgan joyga tushadi. Old devor ochiq. Javob: yo’q.

4-MISOL · QAYSILARI KUB YOYILMASI?



P. Qatorda 4 ta kvadrat — ular tub, o'ng devor, qopqoq, chap devor bo'lib aylanadi. Qolgan ikkitasi qatorning turli tomonida: biri orqa, biri old devor. Kub yoyilmasi.

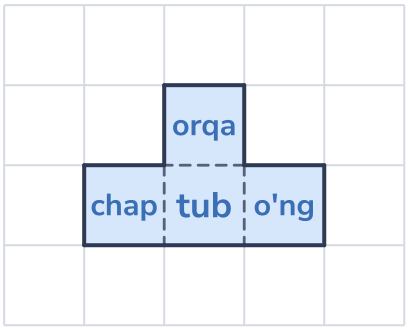
Q. Chap tomonda 4 ta kvadrat «deraza» bo'lib (2×2) turibdi. Pastki chap kvadratni tub qilsak, uning ustidagi ham, o'ngidagining ustidagi ham orqa devorga tushadi. Emas.

R. Ustundagi uchta kvadratning o'rtasi — tub, uning ustidagisi — orqa, ostidagisi — old devor. Chapdagi ikkitasi — chap devor va qopqoq, o'ngdagisi — o'ng devor. Kub yoyilmasi.

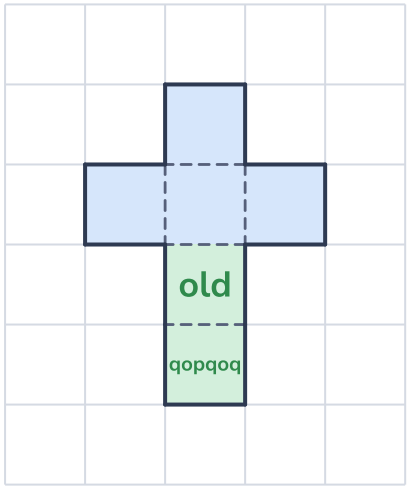
S. Qatordagi 4 ta kvadratning bir tomonida ikkita kvadrat bor. Ikkalasi ham orqa devorga tushadi, old devor ochiq qoladi. Emas.

5-MISOL · CHALA YOYILMANI TO'LDIRISH

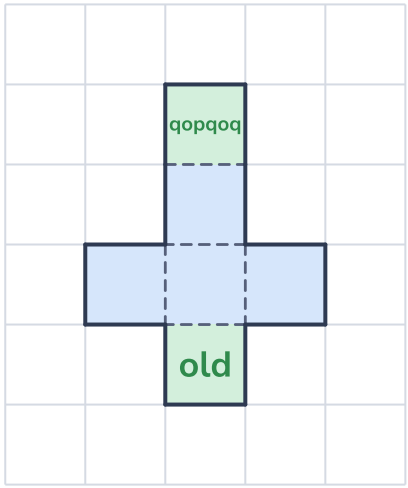
Katakli qog'ozga 4 ta kvadrat chizilgan. Yana 2 ta kvadrat qo'shib, kub yoyilmasini hosil qiling.



1. O'rtadagi kvadrat — tub. Chapdagisi — chap devor, o'ngdagisi — o'ng devor, tepadagisi — orqa devor.
2. Yetishmaydi: old devor va qopqoq. Old devorni tubning ostiga qo'yamiz.
3. Qopqoqni old devorning ostiga yoki orqa devorning ustiga qo'yish mumkin. Masalan, ikki xil javob (boshqa joylari ham bor):



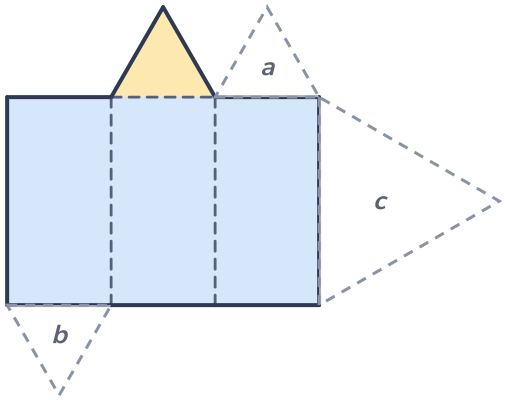
1-javob



2-javob

6-MISOL · PRIZMA YOYILMASINI TO'LDIRISH

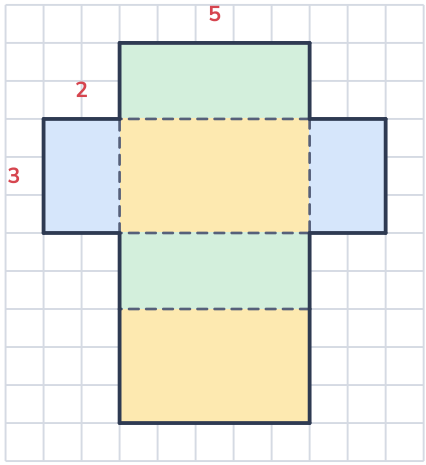
Prizma yoyilmasiga yana bitta uchburchak kerak. a, b, c uchburchaklardan qaysi biri to'g'ri keladi?



1. Tepadagi uchburchak «quvur»ning yuqori uchini yopadi. Demak, ikkinchi uchburchak pastki uchni yopishi kerak.
2. a — yana tepada: ikkita uchburchak bir uchda bo'lib qoladi. c — to'rtburchakning uzun tomoniga yopishgan, u asosdan katta.
3. b pastda turibdi va o'lchami tepadagi uchburchakka teng. Javob: b.

7-MISOL · KUBOID YOYILMASINI KATAKLI QOG'OZGA CHIZISH

Bo'yi 5 birlik, eni 2 birlik, balandligi 3 birlik bo'lgan kuboidning yoyilmasini chizing.

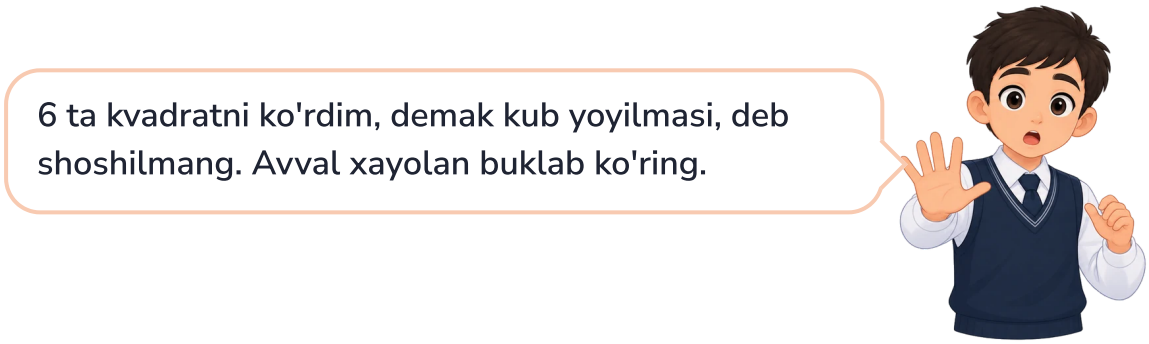


1. Ostki yoqni chizamiz: 5×2 katak.
2. Uning ostiga old yoq — 5×3, keyin ustki yoq — 5×2, keyin orqa yoq — 5×3.
3. Old yoqning ikki yoniga yon yoqlar — 2×3 dan. Jami 6 ta yoq, 3 juft teng.

8-MISOL · YOQLAR, UCHLAR VA QIRRALAR

1-qismdagi 5-misolda ko'rgan prizmaning yoqlari (Y), uchlari (U) va qirralari (Q) sonini toping. $Y + U - Q$ ni hisoblang.

1. Yoyilmada 5 ta shakl — prizmaning yoqlari: $Y = 5$.
2. Har asosda 3 ta uch: $U = 3 + 3 = 6$.
3. Har asosda 3 ta qirra va asoslarni tutashtiruvchi 3 ta qirra: $Q = 3 + 3 + 3 = 9$.
4. $Y + U - Q = 5 + 6 - 9 = 2$.



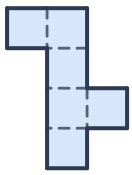
DIQQAT!

1. Kvadratlar «deraza» bo'lib (2×2) tursa yoki 5 tasi bir qatorda tursa — kub yoyilmasi bo'lmaydi.
2. Qatordagi 4 ta kvadratga qolgan ikkitasi turli tomondan yopishadi: ✗ ikkalasi tepada ✓ **biri tepada, biri pastda**
3. Kuboid yoyilmasida teng yoqlar 3 juft bo'ladi, tutashadigan tomonlar teng: ✗ yon yoqlar 2×1 va 2×2 ✓ **ikkala yon yoq 2×1**
4. Prizma yoyilmasida uchburchaklar «quvur»ning ikki uchida bo'ladi: ✗ ikkalasi bir uchida ✓ **biri bir uchida, biri ikkinchi uchida**
5. Piramida yoyilmasidagi yon yoqlar soni asos tomonlari soniga teng: ✗ kvadrat va 3 ta uchburchak ✓ **kvadrat va 4 ta uchburchak**

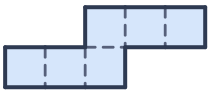
3-QISM

Mustahkamlash

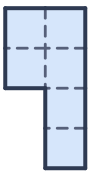
1. a) Qaysi shakllarni buklab kub yasash mumkin? b) Qolgan shakllardan nega kub chiqmaydi?



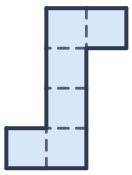
A



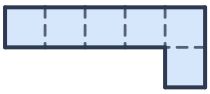
B



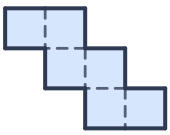
C



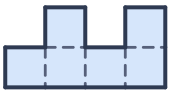
D



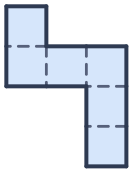
E



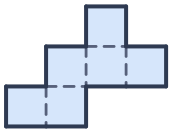
F



G

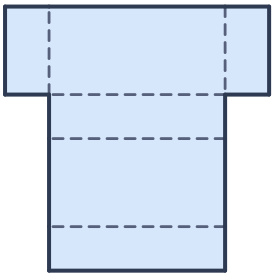


H

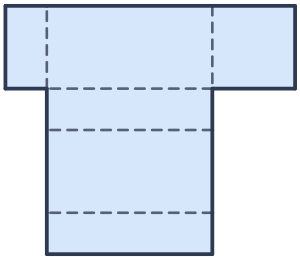


I

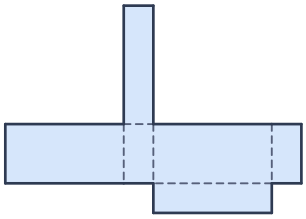
2. Kuboidning bo'yi 4 cm, eni 2 cm, balandligi 1 cm. Shu kuboidni qaysi yoyilmalardan yasash mumkin?



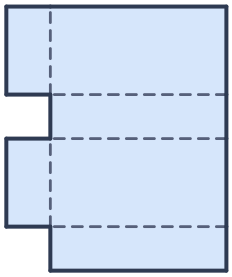
K



L

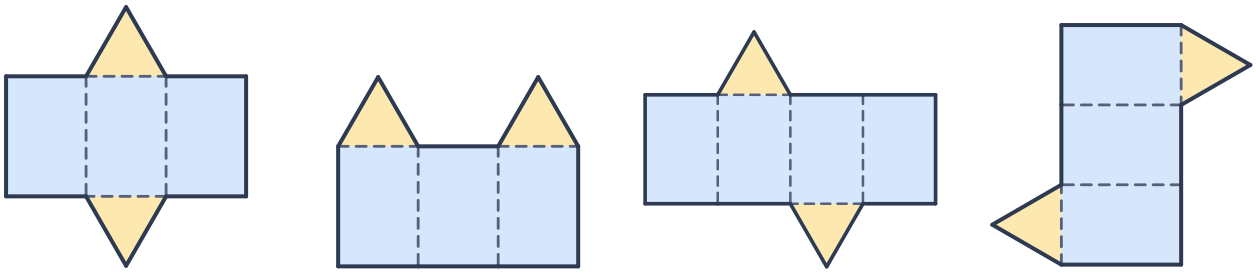


M



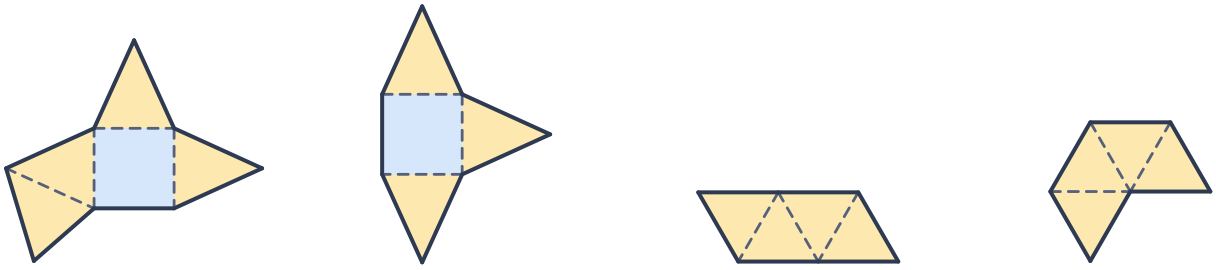
N

3. Qaysi yoyilmalarni buklasak, uchburchakli prizma chiqadi?



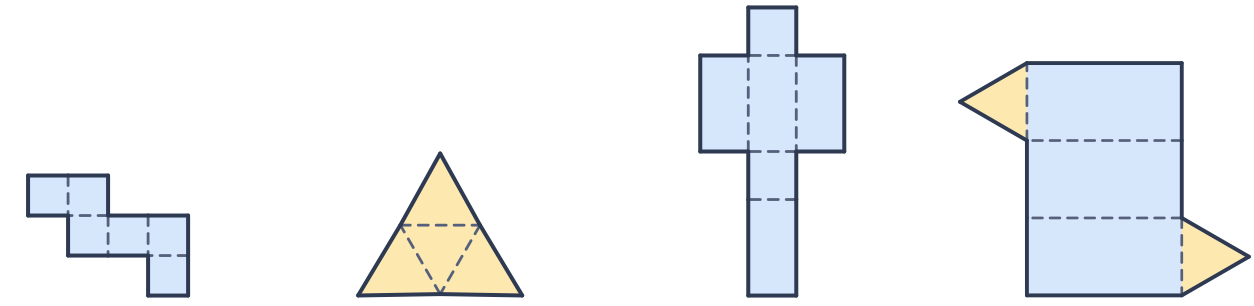
1 2 3 4

4. Qaysi yoyilmalardan piramida yasaladi? (1 va 2 da asos — kvadrat, 3 va 4 da hamma yoqlar teng uchburchaklar.)

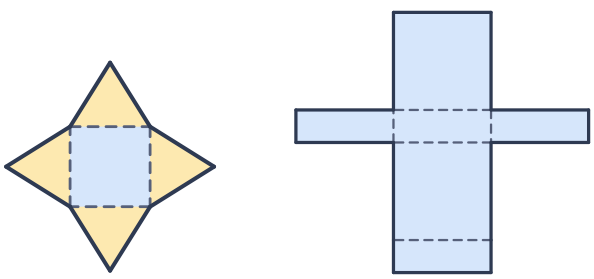


1 2 3 4

5. Har bir yoyilmani buklasak, qanday jism chiqadi?

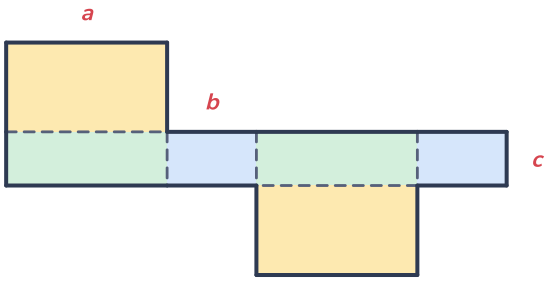


a) b) c) d)

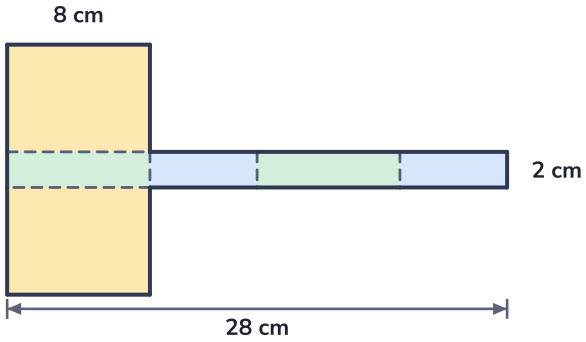


e) f)

6. Kuboidning bo'yi 9 cm, eni 3 cm, balandligi 5 cm. Yoyilmadagi a, b, c kesmalarning uzunligini toping.

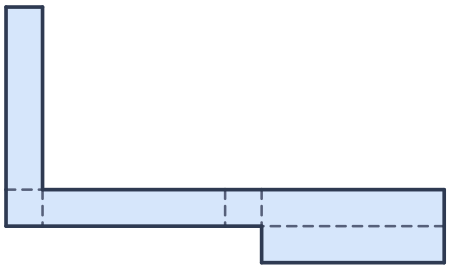


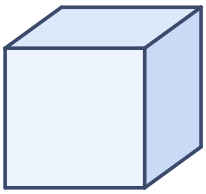
7. Yoyilmaga qarab kuboidning uchta o'lchamini toping.



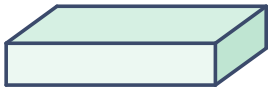
8. Yoyilmaga mos jismni tanlang.

a)





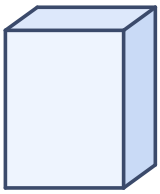
1)



2)

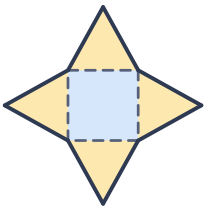


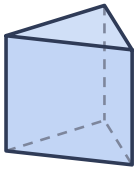
3)



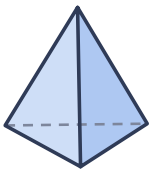
4)

b)

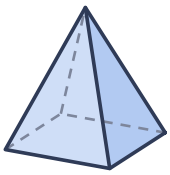




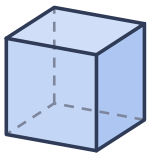
1)



2)

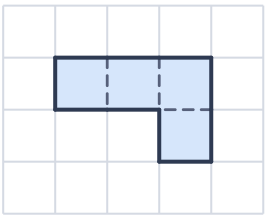


3)

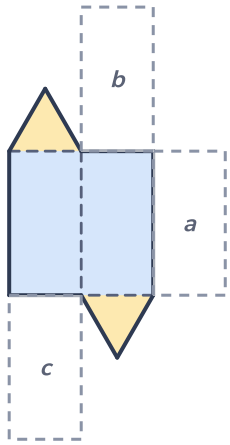


4)

9. Chizmadagi 4 ta kvadratga yana 2 ta kvadrat qo'shib, kub yoyilmasini hosil qiling. Uch xil yo'lini toping.



10. Prizma yoyilmasiga yana bitta to'g'ri to'rtburchak kerak. a, b, c dan qaysi biri to'g'ri keladi?



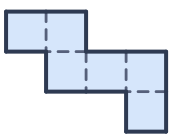
11. Jadvalni daftaringizga chizib to'ldiring. Oxirgi ustunda nimani payqadingiz?

Jism	Yoqlari (Y)	Uchlari (U)	Qirralari (Q)	$Y + U - Q$
Kub				
Kuboid				
Asosi kvadrat piramida				
Asosi uchburchak piramida				

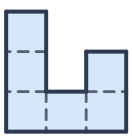
4-QISM

Mustaqil ish

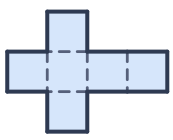
1. Qaysilari kub yoyilmasi? Qolganlari nima uchun emas?



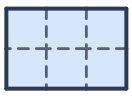
a)



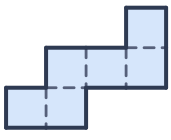
b)



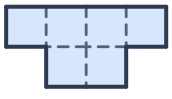
c)



d)

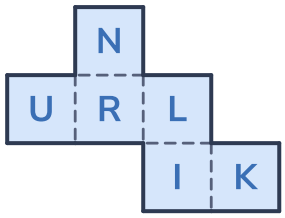


e)

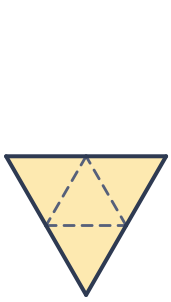


f)

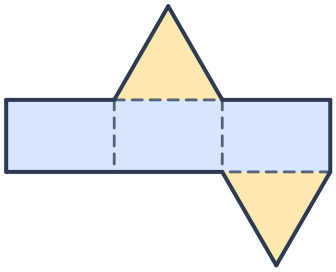
2. Kub yoyilmasining yoqlariga harflar yozilgan. Buklangandan keyin qaysi harflar qarama-qarshi yoqlarda turadi?



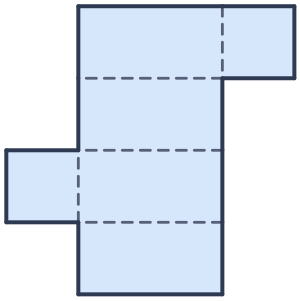
3. Yoyilmalardan qanday jismlar hosil bo'ladi?



a)

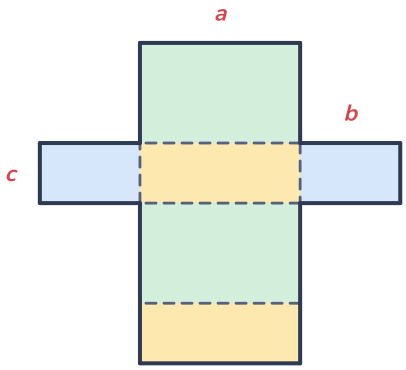


b)

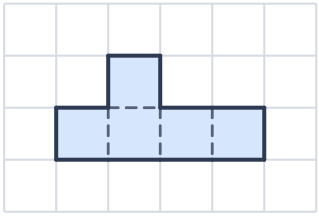


c)

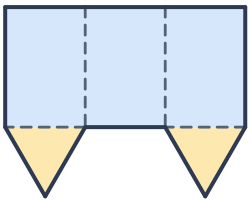
4. Kuboidning bo'yi 8 cm, eni 5 cm, balandligi 3 cm. Yoyilmadagi a, b, c kesmalarni toping.



5. Katakli qog'ozda 5 ta kvadrat bor. Kub yoyilmasi hosil bo'lishi uchun oltinchi kvadratni qayerga qo'yish mumkin? Nechta joy bor?



6. Bu shakl uchburchakli prizmaning yoyilmasimi? Bo'lmasa, bitta uchburchakni qayerga ko'chirish kerak?



7. Bo'yi 4 birlik, eni 2 birlik, balandligi 1 birlik bo'lgan kuboidning yoyilmasini katakli qog'ozga chizing.

8. Asosi kvadrat bo'lgan piramidaning yoyilmasida nechta shakl bor? Piramidaning yoqlari, uchlari va qirralari sonini toping, $Y + U - Q$ ni hisoblang.

9. **O'ylab ko'ring.** Asosi uchburchak bo'lgan yog'och piramidaning uchini asosiga parallel qilib arralab oldik. Arralangan kichik bo'lak qanday jism? Qolgan katta bo'lakni prizma deyish mumkinmi? Nega?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) A, B, D, F, I. b) C — kvadratlar «deraza» (2×2) bo'lib turibdi; E — 5 ta kvadrat bir qatorda, 5-si 1-si bilan ustma-ust tushadi; G — qatordagi 4 ta kvadratning bir tomonida ikkita kvadrat, ikkalasi orqa devorga tushadi; H — qatordagi uchta kvadratning o'rtasini tub desak, chapdagisi chap devor; o'ngdan pastga tushgan ikkinchi kvadrat ham chap devorga tushadi.

2-masala

K va M. L — bitta yon yoq 2×2, u 2×1 bo'lishi kerak (yon yoqlar teng emas). N — ikkala yon yoq chap tomonda: ular ustma-ust tushadi, o'ng tomon ochiq qoladi.

3-masala

1 va 4. 2 da ikkala uchburchak bir uchda (tepada), pastki uch ochiq. 3 da 4 ta to'g'ri to'rtburchak, uchburchakli prizmada esa 3 ta yon yoq bor.

4-masala

1 va 3. 2 da 3 ta uchburchak, kvadrat asosli piramidada esa 4 ta yon yoq kerak. 4 da 4 ta uchburchak bitta uch atrofida: buklasak, asosi kvadrat piramidaning yon yoqlari chiqadi, asosi esa yo'q. Hamma yog'i uchburchak piramidaning har bir uchida 3 ta yoq uchrashadi.

5-masala

a) kub; b) asosi uchburchak bo'lgan piramida; c) kuboid (2 ta teng kvadrat va 4 ta teng to'g'ri to'rtburchak); d) uchburchakli prizma; e) asosi kvadrat bo'lgan piramida; f) kuboid.

6-masala

a — sariq yoqning uzun tomoni: 9 cm. b — ko'k yoqning yuqori tomoni (balandligi): 5 cm. c — ko'k yoqning qisqa tomoni (eni): 3 cm.

7-masala

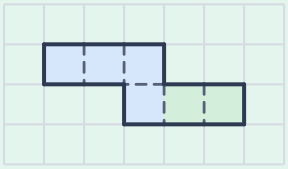
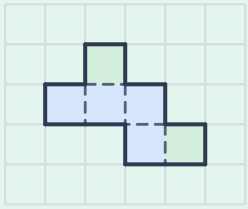
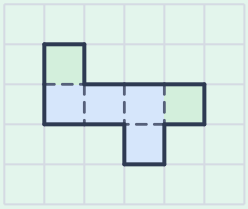
Qatordagi to'rtta yoq: ikkitasi 8 cm, ikkitasi noma'lum. $28 - 8 - 8 = 12$, $12 : 2 = 6$ cm. Qatorning eni 2 cm. O'lchamlar: 8 cm, 6 cm, 2 cm.

8-masala

a) 3) — 4 ta uzun yoq va 2 ta kichik kvadrat: uzun ingichka kuboid. b) 3) — 1 ta kvadrat va 4 ta uchburchak: asosi kvadrat bo'lgan piramida.

9-masala

Masalan (yashil — qo'shilgan kvadratlar):

Boshqa to'g'ri javoblar ham bor — xayolan buklab tekshiring.

10-masala

a. Uchburchaklar allaqachon ikki uchda turibdi, yetishmayotgani — uchinchi yon yoq. U ikkinchi to'rtburchakning uzun tomoniga yopishib, «quvur»ni yopadi. b va c to'rtburchaklar quvurning uchiga yopishgan — u yerga uchburchak kerak.

11-masala

Kub: 6, 8, 12 → 2. Kuboid: 6, 8, 12 → 2. Asosi kvadrat piramida: 5, 5, 8 → 2. Asosi uchburchak piramida: 4, 4, 6 → 2. Hamma jismda $Y + U - Q = 2$.

Mustaqil ish

1-masala

a, c, e — kub yoyilmasi. b — chap ustunning o'rtasini tub desak, o'ng ustunning yuqori kvadrati ham tubga tushadi. d va f — kvadratlar «deraza» (2×2) bo'lib turibdi.

2-masala

L va U, I va N, K va R.

3-masala

a) asosi uchburchak bo'lgan piramida (hamma yoqlari teng uchburchaklar); b) uchburchakli prizma; c) kuboid.

4-masala

a = 8 cm (bo'yi), b = 5 cm (eni), c = 3 cm (balandligi).

5-masala

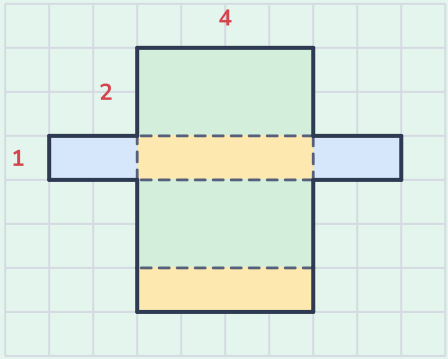
Qatordagi 4 ta kvadratning ostiga — istalgan kvadratning tagiga. 4 ta joy bor. Ustiga qo'ysak, ikkita kvadrat bir tomonda bo'lib qoladi.

6-masala

Yo'q: ikkala uchburchak pastda, yuqori uch ochiq qoladi. Uchburchaklardan birini istalgan to'g'ri to'rtburchakning tepasiga ko'chirish kerak.

7-masala

Masalan:



8-masala

5 ta shakl: 1 ta kvadrat va 4 ta uchburchak. $Y = 5, U = 4 + 1 = 5, Q = 4 + 4 = 8$. Demak, $5 + 5 - 8 = 2$.

9-masala

Kichik bo'lak — piramida: asosi uchburchak, yon yoqlari uchburchaklar, uchi bitta. Katta bo'lak prizma emas: uning ikki asosi (uchburchaklar) teng emas — pastkisi katta, yon yoqlari esa to'g'ri to'rtburchak emas.